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Facile Preparation of 4-(4-Nitrophenyl)morpholin-3-one via the Acid-Catalyzed Selective Oxidation of 4-(4-Nitrophenyl)morpholine by Sodium Chlorite as the Sole Oxidant

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Cite This: *Org. Process Res. Dev.* 2020, 24, 2633−2638 Read Online

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ABSTRACT: 4-(4-Nitrophenyl)morpholin-3-one and 4-(4-aminophenyl) morpholin-3-one are the key intermediates for rivaroxaban synthesis. A facile and economically efficient process has been developed for the preparation of these intermediates. Excellent yield of 4-(4-nitrophenyl)morpholine is obtained by condensing 4-chloro nitrobenzene and morpholine, and 4-(4-nitrophenyl)morpholine is oxidized using inexpensive sodium chlorite to achieve a good yield of the corresponding 4-(4-nitrophenyl)morpholin-3-one. Finally, the key intermediate of rivaroxaban, 4-(4-aminophenyl) morpholin-3-one, is achieved by the iron(III)-catalyzed reduction of the nitro group with aqueous hydrazine. No high-cost materials were used, and the process did not require column purification.

KEYWORDS: *rivaroxaban, 4-(4-nitrophenyl)morpholin-3-one, oxidation, sodium chlorite*

INTRODUCTION

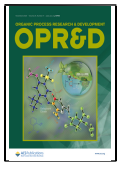
4-(4-Nitrophenyl)morpholin-3-one (**1**) and 4-(4-aminophenyl) morpholin-3-one (**2**) are the key intermediates in the preparation of rivaroxaban (marketed as XARELTO), an oxazolidinone derivative anticoagulant, which displays remarkable therapeutic effects in the management of several diseases.[1] (Scheme 1)

Intermediate **1** can be transformed into the corresponding intermediate **2** easily via a nitro reduction reaction, and compound **2** is usually used as the starting material for the preparation of rivaroxaban.[2] Several methods for the synthesis of compound **1** have been reported. Bayer AG reported the condensation of 3-morpholinone with 1-fluoro-4-nitrobenzene in the presence of *t*-BuOK or NaH in NMP to give **1** with a lower yield (17.4%).[3] An alternative route was applied to achieve precursor **2** directly by coupling 3-morpholinone with 4-halogen-aniline (4-bromo aniline or 4-iodo aniline) with a good yield.[2a] However, in both cases, 3-morpholinone is the key material, but it is difficult to obtain due to its low yield,[4] thereby increasing the cost of the whole process (Scheme 2).

Other routes to obtain intermediates **1** and **2** have also been reported. The coupling of bromobenzene with 2-amino ethanol yielded 2-phenylamino ethanol, which was condensed with methyl 2-chloro acetate, and subsequently nitrated to achieve **1**. Alternatively, compound **1** was also prepared by the condensation of 4-bromo nitrobenzene with 2-amino ethanol, followed by cyclization with methyl 2-chloro acetate. Both methods displayed low efficiency because of two reactive sites existing in both reactants, resulting in a complicated reaction system[5] (Scheme 3).

Several processes have been reported for the transformation of cyclic amines into corresponding lactams via the chemoselective and regioselective oxidation of C−H bonds directly adjacent to a cyclic amine. In 2009, a patent showed an oxidation protocol for preparing compound **1** (Scheme 4, eq 1).[3] In this case, large amounts of heavy-metal oxidants, $KMnO_4$, were utilized; however, this protocol is not environment friendly and is thus unfit for industry use. In 2014, Milstein et al. showed that cyclic amines can be transformed into lactams via ruthenium-catalyzed dehydrogenation (Scheme 4, eq 2); however, completing this transformation involves harsh reaction conditions and an air-sensitive and precious metal catalyst.[6] Ferric chloride-catalyzed oxidation cyclic amines offered a lower-cost process for obtaining lactams (Scheme 4, eq 3), but the substrate scope is currently limited due to the use of a strong peroxide oxidant.[7] The heterogeneous gold nanoparticle supported on alumina is a mild and chemoselective route for amide and lactam formation (Scheme 4, eq 4);[8] however, the catalyst loading is too large, the gold catalyst is costly, and the catalyst preparation is tedious. From an environmental perspective, the transient metal-free process is promising. In 2017, Talbot et al. developed a metal-free, chemoselective, and regioselective lactam formation method using molecular iodine as an oxidant to oxidize cyclic amine (Scheme 4, eq 5).[9] However, this process uses a large amount of iodine and sodium bicarbonate, which produces too much waste. In 2018, Sartillo-Piscil et al. reported 2,2,6,6-tetramethylpiperidin-1-yloxy (TEMPO)-catalyzed oxidation of piperazines and morpholines under transition-metal-free conditions (Scheme 4, eq 6).[10] This

Received: June 29, 2020
Published: September 14, 2020

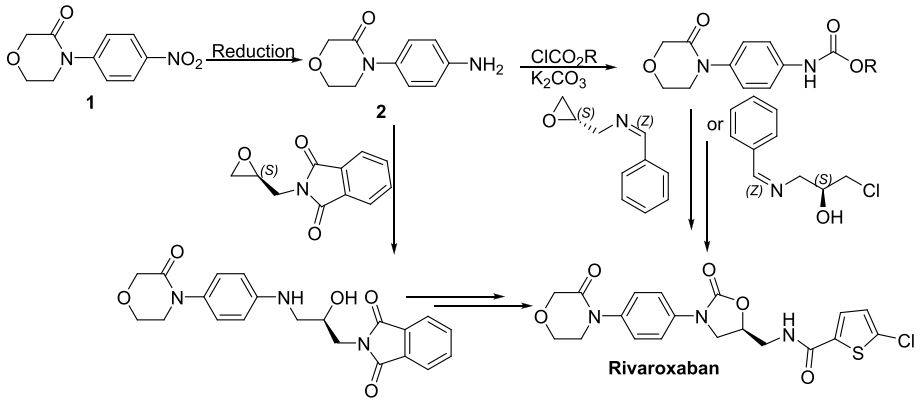


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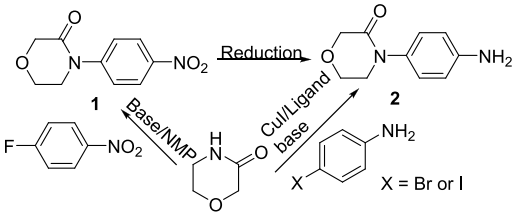
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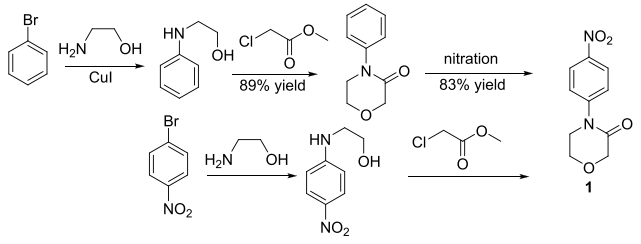
Scheme 1. Synthetic Routes of Rivaroxaban



Scheme 2. Synthetic Routes of Intermediates 1 and 2



Scheme 3. Alternative Synthetic Routes of Intermediate 1



protocol uses low-cost and environmentally friendly oxidants, including sodium chlorite and sodium hypochlorite. Herein, we applied this catalytic reaction system to transform 4-(4-nitrophenyl)morpholine into 4-(4-nitrophenyl)morpholin-3-one **1**.

However, use of a TEMPO-catalyzed oxidation system in industrial processes has obvious disadvantages, as it involves a high-cost TEMPO catalyst and produces high amounts of waste salts. Furthermore, the complex reaction system would cause reaction control problems. The mixture of sodium chlorite and bleach appears unstable, which may lead to spray incidents.[11] As such, we describe here a novel acid-catalyzed oxidation of 4-(4-nitrophenyl) morpholine into **1** under mild conditions, by using the low-cost and environmentally friendly sodium chlorite as the sole oxidant (Scheme 5).

RESULTS AND DISCUSSION

4-(4-Nitrophenyl)morpholine (**3**) is prepared via the condensation of 4-chloro nitrobenzene with morpholine under neat conditions. Other 4-halogen nitrobenzenes, such as 4-bromo-nitrobenzene, 4-fluoro-nitrobenzene, and 4-iodo-nitrobenzene, are suitable for preparing 4-(4-nitrophenyl)-morpholine.[12] In the next step, we explored the oxidation of 4-(4-nitrophenyl) morpholine (Table 1). When the oxidation reaction was performed according to the reference reaction condition,[10] 98% of the substrate (**3**) was consumed. Aside from the generated target product 4-(4-nitrophenyl) morpholine-3-one, two side products (**1a** and **1b**) were detected, suggesting that a side chlorination reaction also occurred (Table 1, entry 1). Further increasing the reaction temperature (from 0 °C to room temperature) expedited this oxidation reaction dramatically with almost the same regioselectivity (Table 1, entry 2). Given that TEMPO is an expensive catalyst, which would be destroyed under reaction conditions to form impurities, we performed the reaction without TEMPO. Almost the same result was achieved (Table 1, entry 3). Further experiment showed that both TEMPO and sodium hypochlorite could cause chloride (**1a**) formation and resulted in better selectivity without using TEMPO and sodium hypochlorite as additives (Table 1, entries 4 and 5). The buffer NaH_2PO_4 is the key species for this reaction because it acts as acid to adjust the reaction system's pH. No reaction was observed when only NaClO or TEMPO was used, as well as without any additives (Table 1, entries 6−8).

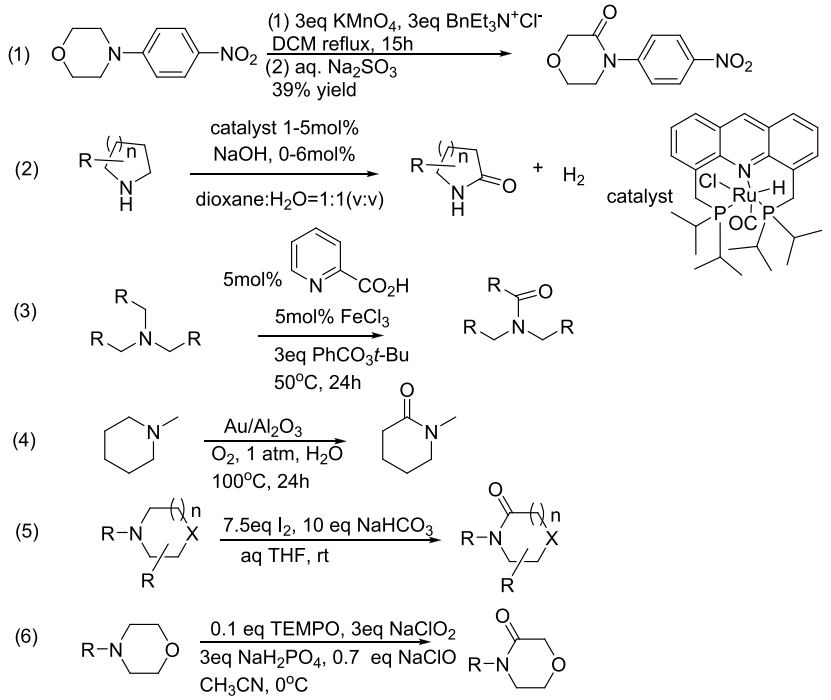
NaH_2PO_4, Na_2HPO_4, and Na_3PO_4 usually act as buffers in most reactions. NaH_2PO_4 was replaced with Na_2HPO_4 and Na_3PO_4 in this oxidation reaction, but no reaction occurred (Table 2, entries 1 and 2). This result may be attributed to the difference in the pH of the reaction system: Na_2HPO_4, 10.26, Na_3PO_4, 12.6, and NaH_2PO_4, 4.12−4.24. Thus, we believe that the acidic condition is critical for this reaction. Furthermore, excessive amounts of NaH_2PO_4 in this reaction would lead to a large amount of solid waste. Thus, several acid additives were screened for this cyclic amine oxidation reaction. This reaction was completed in 45 min with low selectivity using 1.5 equiv of acetic acid (Table 2, entry 3). Decreasing the amount of acetic acid can increase selectivity (Table 2, entries 4 and 5), and 0.3 equiv of acetic acid resulted in the best conversion and selectivity in 4 h (Table 2, entry 6). HPLC indicated that the reaction with 0.2 equiv of acetic acid was only completed after 4 h, suggesting that the reaction was slow overall (Table 2, entry 7). Other acids, such as formic acid and phosphoric acid, could also promote this reaction to consume all starting materials (Table 2, entries 8 and 9), but an oxidation reaction with formic acid as an additive showed poor selectivity. The high reaction temperature would accelerate the reaction rate, and the reaction could be completed in 2 h at 50 °C in the presence of 0.3 equiv of acetic acid (pH 5.23−7.42) with good inversion rate and selectivity (Table 2, entry 10).

Finally, the amount of oxidant was screened (Table 3). From the results, 2 equiv of sodium chlorite was enough to consume all substrates; however, both reaction rates and selectivity were low (4 h, 83%) (Table 3, entry 1). Increasing the amount of

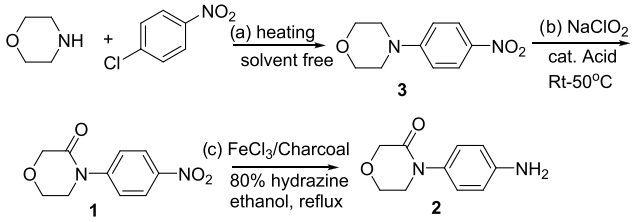
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Scheme 4. Processes for the Transformation of Amines into Their Corresponding Amides via Chemoselective and Regioselective Oxidation



Scheme 5. Oxidation Route for the Preparation of Intermediates 1 and 2[a]



[a]Reaction conditions: a, morpholine (1 mol), 4-chloro nitrobenzene (0.2 mol), Na_2CO_3 (0.12 mol), 100 °C, 4.5 h; b, $NaClO_2$ (300 mol %), acetic acid (30% mol); and c, $FeCl_3$/charcoal(15 mol %), aq. hydrazine (500 mol %), and reflux.

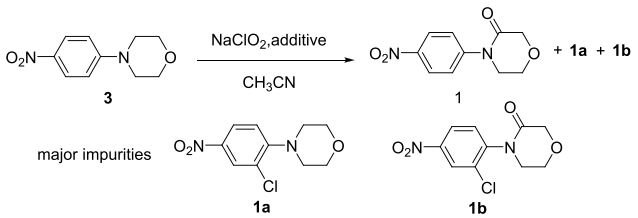
oxidant could enhance selectivity (Table 3, entry 2), and more than 3 equiv of sodium chlorite resulted in the best selectivity (Table 3, entries 3 and 4).

With the optimized reaction condition (3 equiv of sodium chlorite and 0.3 equiv of acetic acid in acetonitrile), the oxidation reaction profile was established using the HPLC yield (**1**/**1a**/**1b**). As shown in Figure 1, the oxidation reaction rate is dramatically faster than the chlorination reaction rate. The reaction was completed in 2.5 h, and the total amounts of **1a** and **1b** were not obviously increased.

Furthermore, **1a** can be prepared by chlorinating **3** with sodium hypochlorite under acidic conditions with a good yield. When **1a** is loaded in this oxidation reaction system, it can be oxidized smoothly to the corresponding **1b** with an 86% yield (HPLC purity, 98%). The control experiment shows that **1** cannot be chlorinated to **1b** with sodium hypochlorite under acidic conditions.

With the easily prepared intermediate **1**, the reduction of the nitro group in **1** was also investigated (Scheme 6). The known method uses Pd on charcoal to catalyze hydrogenation.

Table 1. Critical Parameters of the Oxidation Reaction[a]



	additive (mol)					conv. (%)[b]	molar ratio[b]
entry	NaH_2PO_4	NaClO	TEMPO	temp. (°C)	T (h)		1:1a:1b
1	3 equiv	0.7 equiv	0.1 equiv	0−r.t.	18	>98	86:7:6
2	3 equiv	0.7 equiv	0.1 equiv	25	9	>98	88:7:4
3	3 equiv	0.7 equiv	0	25	10.5	>98	86:8:5
4	3 equiv	0	0.1 equiv	25	10	>98	90:4:5
5	3 equiv	0	0	25	3	>99	94:3:2
6	0	0.7 equiv	0	25	3	0	nr
7	0	0	0.1 equiv	25	3	0	nr
8	0	0	0	25	3	0	nr

[a]Reaction conditions: unless specified, a mixture of **2** (5 mmol), $NaClO_2$ (15 mmol) in water (6 mL), and CH_3CN (15 mL) was stirred. [b]Conversions and yields based on high-performance liquid chromatography (HPLC) (area normalization).

However, a large amount of catalyst (5−20 wt %, 5% palladium on charcoal) is needed, thereby leading to higher production cost.[13] We used iron(III)−charcoal-catalyzed reduction of the nitro group with aqueous hydrazine as the reductant; however, a one-pot reaction resulted in the

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Table 2. Screening Additive Acids in Amine Oxidation Reactions

entry	additive (mol)	T (h)	conv. (%)[b]	molar ratio[b] 1:1a:1b
1	Na_2HPO_4 (3 equiv)	overnight	0	nr
2	Na_3PO_4 (3 equiv)	overnight	0	nr
3	CH_3CO_2H (1.5 equiv)	0.75 h	>99	82:12:6
4	CH_3CO_2H (1 equiv)	1 h	>99	87:9:4
5	CH_3CO_2H (0.5 equiv)	3 h	>99	91:6:3
6	CH_3CO_2H (0.3 equiv)	4 h	>99	94:3:2
7	CH_3CO_2H (0.2 equiv)	7 h	>99	90:5:3
8	HCO_2H (0.3 equiv)	5 h	96	89:3:3
9	H_3PO_4 (0.2 equiv)	2 h	>99	96:3: 1
10	CH_3CO_2H(0.3 equiv)[a]	2 h	>99	96:2:2

[a]All reactions were performed at 50 °C. [b]determined by area normalization on HPLC.

Table 3. Screening the Amount of Sodium Chlorite[a]

entry	amount of sodium chlorite (mol)	time (h)	molar ratio[b] 1:1a:1b
1	2 equiv	4.0	83:14:3
2	2.5 equiv	3.0	90:8:2
3	3 equiv	2.0	96:2:2
4	4 equiv	1.5	97:2:1

[a]Reaction conditions: unless specified, a mixture of 2 (5 mmol) and CH_3CO_2H (1.5 mmol) was dissolved in CH_3CN (10 mL) and stirred at 50 °C; then, $NaClO_2$ in water (3.5 mL) was added dropwise. [b]determined by area normalization on HPLC.

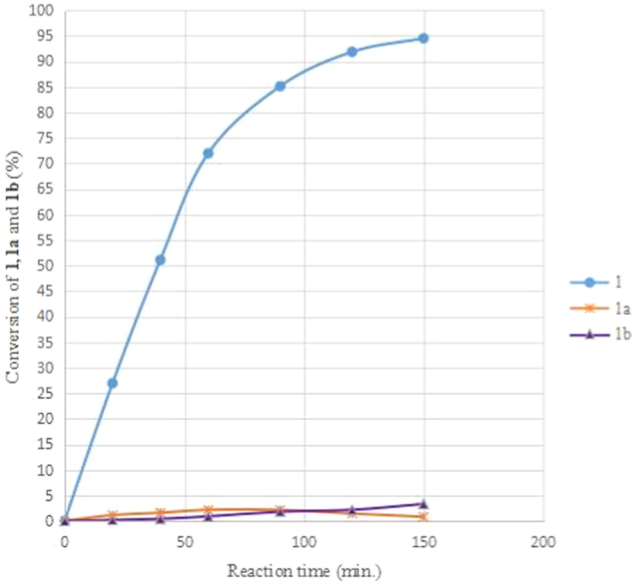


Figure 1. HPLC yield (**1**/**1a**/**1b**) with reaction time.

morpholin-3-one ring-opening compound **2a** as the main product. In the iron(III)−hydrazine reduction system, the real active species is diimide, which is formed via in situ oxidation of hydrazine by iron(III). Then, hydrazine was added dropwise to the reaction, and the side product was dramatically decreased.

We have developed a facile and convenient process of preparing 4-(4-nitrophenyl) morpholin-3-one, which can be further reduced to its corresponding 4-(4-aminophenyl) morpholin-3-one. Our process involves low-cost and easily available materials. Furthermore, working up the reaction in the whole process is quite convenient, and column chromatography is not needed. The product can be obtained via a simple slurry and filtration and is suitable for industrial applications.

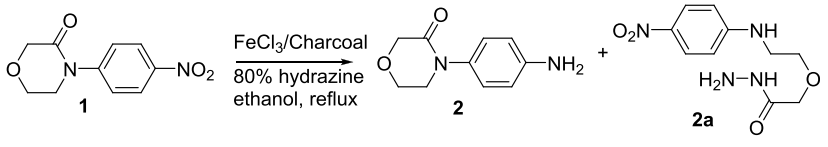
EXPERIMENTAL SECTION

General Procedures. Reverse-phase high performance liquid chromatography (RP-HPLC) was performed on an Agilent 1260 series HPLC instrument using the following methods. Zorbax sb-c18 RP (150 mm × 4.6 mm, 3.5 μ); conditions: flow rate 1 mL/min; 30 °C; mobile phase: (A) water, (B) acetonitrile, and (C) methanol; gradient: 0−4 min, 65−10% A, 25−80% B, 10% C; 4−7 min, 10% A, 80% B, 10% C; 7−14 min, 10−65% A, 80−25% B, 10% C; 14−16 min, 65% A, 25% B, 10% C; and 230 nm UV detector. Retention times are uncorrected; molar ratios are determined by area normalization.

*4-Nitrophenyl Morpholine (**3**).* A mixture of *p*-nitrochlorobenzene (31.5 g, 0.2 mol), morpholine (87 g, 1 mol), and sodium carbonate (12.7 g, 0.12 mol) in a 500 mL round-bottom flask was stirred at 100 °C for 4.5 h; TLC and HPLC detection showed that *p*-nitrochlorobenzene was almost consumed. Then, most of the unreacted morpholine was recovered by vacuum distillation. After that, 200 mL of water was added to this residue and stirred for a few moments. All solid products were collected by filtration, rinsed with 50 mL of water, and dried under vacuum to afford 4-nitrophenyl morpholine (**3**). Yellow solid (isolated yield: 99%); ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.22−8.13 (m, 2H), 6.93−6.84 (m, 2H), 3.95−3.86 (m, 4H), 3.44−3.36 (m, 4H); ^{13}C NMR (101 MHz, $CDCl_3$) δ 154.77, 139.25, 125.87, 112.90, 66.28, 47.32. MS (ES-API) $[M + 1]^+$ = 209.

*4-(4-Nitrophenyl)morpholin-3-one (**1**).* A solution of 4-nitrophenyl morpholine (20.8 g, 0.1 mol) and acetic acid (1.8 g, 0.03 mol) in acetonitrile (200 mL) was added into a 1000 mL three-port round-bottom flask, and it was heated at 50°C. Then, a solution of sodium chlorite (more than 80%, 33.8 g, 0.3 mol) in 65 mL of water was added dropwise over 20 min. The reaction system gradually turned brown red and then light yellow. TLC and HPLC showed that the reaction was completed in 2.5 h. The reaction was quenched with aqueous saturated sodium sulfite, and the organic layer was separated. Most organic solvent was removed by evaporation, then the solid was collected by filtration, rinsed with a small amount of

Scheme 6. $FeCl_3$-Catalyzed Aqueous Hydrazine Reduction of Intermediate 1 to 2



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water, and dried; it could be used in the next step without further purification. **1**, Light yellow solid (isolated yield: 91%, purity 99%, HPLC);^{1}H NMR (400 MHz,$CDCl_3$) δ 8.28 (d, J = 8.8Hz, 2H), 7.61 (d, J = 8.8Hz, 2H), 4.40 (s, 2H), 4.08 (t, J = 4.8Hz, 2H), 3.85 (t, J = 4.8Hz, 2H); ^{13}C NMR (101 MHz,$CDCl_3$) δ 166.79, 146.73, 145.40, 124.69, 124.54, 68.64, 63.94, 48.87; MS (ES-API) $[M + 1]^+$ = 223. Side products **1a** and **1b** were isolated from the mother liquid, and purified on a silica gel (petroleum ether/ ethyl acetate as an eluent).

4-(2-Chloro-4-nitrophenyl)morpholine (1a). Yellow solid (isolated 120 mg, yield: 0.5%); ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.26 (d, J = 2.8 Hz, 1H), 8.11 (dd, J = 9.2, 2.8 Hz, 1H), 7.05 (d, J = 9.2 Hz, 1H), 3.89 (t, J = 4.8 Hz, 4H), 3.21 (t, J = 4.8 Hz, 4H); ^{13}C NMR (101 MHz,$CDCl_3$) δ 154.50, 142.43, 127.67, 126.70, 123.45, 119.26, 66.71, 50.99; MS (ES-API) $[M + 1]^+$ = 243.

4-(2-Chloro-4-nitrophenyl)morpholine-3-one (1b). Light yellow solid (isolated 230mg, yield: 0.9%); ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.40 (d, J = 2.4 Hz, 1H), 8.24 (dd, J = 8.4, 2.4Hz, 1H), 7.51 (d, J = 8.4 Hz, 1H), 4.41 (s, 2H), 4.15−4.07 (m, 2H), 3.74 (t, J = 5.0 Hz, 2H); ^{13}C NMR (101 MHz, $CDCl_3$) δ 166.47, 147.64, 144.21, 133.85, 130.45, 126.03, 123.18, 68.45, 63.93, 49.26; MS: $[M + 1]^+$ = 257. After the oxidation reaction was performed for 2.5 h, the reaction mixture was detected by HPLC. The HPLC spectrum can be found in the Supporting Information. Retention times for **1**, **1a**, **1b**, and **3** are 6.2, 8.9, 6.7, and 7.7 min, respectively.

1a could be prepared according to the following procedure: a mixture of 4-nitrophenyl morpholine (1.04 g, 5 mmol) and acetic acid (0.09 g, 1.5 mmol) in 10 mL of acetonitrile was heated to 40°C in a water bath and then sodium hypochlorite (7% activated Cl, 8.34 g, 9 mmol) was added dropwise. After 4-nitrophenyl morpholine was consumed (TLC), the reaction was quenched by adding aqueous saturated sodium sulfite. After most of the solvent was removed by evaporation, a large amount of solid was precipitated. This solid was collected by filtration and dried to obtain **1a** (yield 88%, purity 98%, HPLC).

1b can be prepared from **1a** (1.21 g, 5 mmol) by the same procedure as that for the preparation of **1**. Yield, 86%, HPLC purity 98%.

When the mixture of **1** and acetic acid in acetonitrile was treated with sodium hypochlorite (7% activated Cl), no reaction occurred.

4-(4-Aminophenyl)morpholin-3-one (2). To a solution of 4-(4-nitrophenyl)morpholin-3-one (11.1 g, 0.05 mol) in 150 mL of ethanol, anhydrous ferric chloride (1.215 g, 7.5 mmol) and activated carbon (5 g) were added subsequently. Then, the reaction was heated to reflux, and 15.6 g of hydrazine hydrate (80% aqueous solution, 0.25 mol) was added dropwise over 1 h. TLC and HPLC detection showed that this reaction could be completed in 2h. Charcoal was removed by filtration, and the solution was concentrated on a rotatory evaporator to leave a residue, which was recrystallized from acetonitrile to afford pure **2**. White solid (isolated yield: 93%, purity 98%, HPLC); ^{1}H NMR (400 MHz,DMSO-d_6) δ 6.96 (d, J = 5.6 Hz, 2H), 6.56 (d, J = 5.6 Hz, 2H), 5.13 (s, 2H), 4.14 (s, 2H), 3.91 (t, J = 3.2 Hz, 2H), 3.58 (t, J = 3.2 Hz, 2H); ^{13}C NMR (101 MHz, $CDCl_3$) δ 165.74, 147.31, 130.21, 126.39, 113.65, 67.68, 63.51, 49.58. MS(ES-API) $[M + 1]^+$ = 193.

When the reaction was performed in one pot, a yellow solid was formed; the structure was confirmed to be the ring-opening product **2a**.

2-(2-((4-Nitrophenyl)amino)ethoxy) Acetohydrazide (2a). Yellow solid; ^{1}H NMR (400 MHz, DMSO-d_6) δ 9.04 (s, 1H), 8.00(d, 2H, J = 9.2 Hz), 7.38 (t, J = 5.2 Hz, 1H), 6.70 (d, 2H, J = 9.2 Hz), 4.27 (s, 2H), 3.93 (s, 2H), 3.60 (t, J = 5.2 Hz, 2H), 3.39 (t, J = 5.2 Hz, 2H); ^{13}C NMR (101 MHz, DMSO-d_6) δ 167.98, 154.45, 135.64, 126.17, 110.79, 69.28, 69.01, 42.04. MS(ES-API) $[M + 1]^+$ = 255.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.oprd.0c00299.

Spectra for some compounds (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (21372081, 21172072).

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Org. Process Res. Dev. 2020, 24, 2633−2638